

Program :

use college

output :

switched to db college

```
db.createCollection("Student");
```

output:

```
{ "ok" : 1 }
```

```
db.Student.insertMany([
  { "rollno": 1, "name": "Alice", "marks": 85 },
  { "rollno": 2, "name": "Bob", "marks": 65 },
  { "rollno": 3, "name": "Charlie", "marks": 45 },
  { "rollno": 4, "name": "David", "marks": 75 },
  { "rollno": 5, "name": "Eva", "marks": 35 },
  { "rollno": 6, "name": "Rohan", "marks": 30 }
]);
```

Output :

```
{
  "acknowledged" : true,
  "insertedIds" : [
    ObjectId("68d94f79c2f4efc22410e37e"),
    ObjectId("68d94f79c2f4efc22410e37f"),
    ObjectId("68d94f79c2f4efc22410e380"),
    ObjectId("68d94f79c2f4efc22410e381"),
    ObjectId("68d94f79c2f4efc22410e382"),
    ObjectId("68d94f79c2f4efc22410e383")
  ]
}
```

```
var mapFunction = function() {
  var category = "";
  if (this.marks >= 70) {
    category = "Passed with Distinct Class";
  } else if (this.marks >= 50 && this.marks < 70) {
    category = "Passed with First Class";
  } else if (this.marks >= 40 && this.marks < 50) {
    category = "Passed with Second Class";
  } else {
    category = "Fail";
  }
  emit(category, 1);
};
```

```
var reduceFunction = function(key, values) {  
  var total = 0;  
  values.forEach(function(item) {  
    total += item;  
  });  
  return total;  
};
```

```
db.Student.mapReduce(  
  mapFunction,  
  reduceFunction,  
  {  
    out: "class_counts"  
  }  
);
```

Output :

```
{ "result" : "class_counts", "ok" : 1 }
```

```
db.class_counts.find().pretty();
```

output :

```
{ "_id" : "Passed with Distinct Class", "value" : 2 }  
{ "_id" : "Passed with First Class", "value" : 1 }  
{ "_id" : "Fail", "value" : 2 }  
{ "_id" : "Passed with Second Class", "value" : 1 }
```